ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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- **University: NTHU**

Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. 594
2. Isolating a specific category helps create a task-relevant dataset, so that the patterns discovered by frequent pattern mining are related to the target domain. If documents from unrelated categories are mixed together, frequent patterns may reflect general or irrelevant relationships rather than meaningful patterns within the category of interest.

AI used for this answer: Yes

If yes, how: using AI to search for relevant references and correct grammatical errors in my writing.

External references (links):

https://studylib.net/doc/28135705/concepts-and-techniques--4th-edition-?utm_source=chatgpt.com&p=3

https://www.researchgate.net/publication/220254274_A_survey_of_data_mining_and_knowledge_discovery_process_models_and_methodologies

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Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer: Simply dropping the rows would discard 40% of the documents and their potentially useful text content, causing substantial information loss. Instead, we can use fillna() to assign a placeholder such as "Unknown" to the missing category_name values, allowing the documents to be retained and their text data to still be used for further analysis.

AI used for this answer: Yes

If yes, how: using AI to correct grammatical errors in my writing.

External references (links):

https://pandas.pydata.org/pandas-docs/stable/user_guide/missing_data.html#filling-missing-data

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Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. No. The proportions are similar, but they do not match exactly. For example, soc.religion.christian is the largest category in the original data, while comp.graphics is the largest in the sample.
2. Random sampling may select too few examples of a minority class, or miss it entirely. This makes the sample less representative. Stratified sampling can help preserve the original class proportions.

AI used for this answer: Yes

If yes, how: using AI to search for relevant references and correct grammatical errors in my writing.

External references (links):

https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.sample.html

https://scikit-learn.org/stable/modules/cross_validation.html#stratification

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Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. The number of features increases because stopwords are now included in the vocabulary.
2. The matrix becomes less sparse because common stopwords add many non-zero entries, reducing the percentage of zeros.

AI used for this answer: Yes

If yes, how: using AI to search for relevant references and correct grammatical errors in my writing.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.CountVectorizer.html

https://scikit-learn.org/stable/modules/feature_extraction.html#sparsity

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Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

1. As the threshold increases, fewer terms are kept. The vocabulary decreases from 35,482 terms at 0.0 to 1,491 terms at 0.05.
2. Very rare words are usually removed first because their counts are zero in almost every document. Words with nearly identical counts across documents also have low variance.

3. Near-constant features usually provide little information for distinguishing documents, so removing them can reduce dimensionality with limited impact on accuracy. However, this is not guaranteed, since some rare words may still be useful for identifying a minority class.

Threshold	**Number of terms kept (keep == True)**
0.0	35482
0.01	5089
0.05	1491

AI used for this answer: Yes

If yes, how: using AI to explain variance thresholding, search for relevant references and correct grammatical errors in my writing.

External references (links):

https://scikit-learn.org/stable/modules/feature_selection.html#removing-features-with-low-variance

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Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?

3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. Both lists mix general words with graphics-related terms. Pearson includes “vga,” “video,” and “card,” while Spearman includes “graphics,” “image,” and “3d.” I inspected the helper output, which ranks features by absolute correlation.
2. Pearson measures linear relationships using original values. Spearman measures monotonic relationships using ranks.
3. If a word is usually more frequent in graphics documents but a few posts repeat it excessively, Spearman may stay high while Pearson is lowered by the extreme counts.

AI used for this answer: Yes

If yes, how: using AI to search for relevant references and correct grammatical errors in my writing.

External references (links):

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.pearsonr.html?utm_source=chatgpt.com

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.spearmanr.html?utm_source=chatgpt.com

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Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

Lowering minSup from 9 to 3 increased runtime from about 0.15 to 3.31 seconds and increased the number of maximal patterns from 710 to 9,547. The lower threshold allows less frequent patterns to qualify and requires more searching.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

Top-K FAE uses k to request the k most frequent patterns; my runs returned 100 and 500 patterns. MaxFPGrowth uses minSup as a minimum support requirement and keeps only maximal frequent patterns, so the output count is not fixed in advance.

Answer (3) — Maximal vs standard frequent patterns:

Maximal frequent patterns have no larger frequent superset. Unlike standard frequent-pattern mining, they omit subsets already covered by larger frequent patterns, producing a more compact result.

AI used for this answer: Yes

If yes, how: using AI to clarify the questions and concepts, help troubleshoot errors in the notebook, search for relevant references and correct grammatical errors in my writing.

External references (links):

https://github.com/UdayLab/PAMI/blob/main/notebooks/frequentPattern/topk/FAE.ipynb

https://github.com/UdayLab/PAMI/blob/main/notebooks/frequentPattern/maximal/MaxFPGrowth.ipynb

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Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. Each document has tens of thousands of feature values, but a standard scatter plot has only two or three axes. Therefore, we cannot directly display all dimensions of X_counts in one scatter plot.
2. Dimensionality reduction algorithms, such as PCA, t-SNE, or UMAP, should first map the documents into two or three dimensions for visualization.

AI used for this answer: Yes

If yes, how: using AI to search for relevant references and correct grammatical errors in my writing.

External references (links):

https://scikit-learn.org/stable/modules/decomposition.html#pca

https://scikit-learn.org/stable/modules/manifold.html?utm_source=chatgpt.com

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Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. PCA shows heavy overlap near the origin, with a few distant outliers. t-SNE and UMAP reveal some distinct local clusters, especially among the black points, but substantial category overlap remains. Neither method cleanly separates all four categories.
2. PCA finds linear directions with the greatest variance, which may reflect document length or common words rather than category differences. t-SNE and UMAP can capture nonlinear neighborhood structure, so they may reveal groupings that PCA misses.
3. t-SNE mainly preserves local neighborhoods, not distances between separate clusters. A larger gap between two clusters does not necessarily mean their topics are more different.

AI used for this answer: Yes

If yes, how: using AI to clarify the questions and concepts, help troubleshoot errors in the notebook, search for relevant references and correct grammatical errors in my writing.

External references (links):

https://scikit-learn.org/stable/modules/decomposition.html#pca

https://scikit-learn.org/stable/modules/manifold.html#t-sne

https://distill.pub/2016/misread-tsne/

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. Four binary columns are created, one for each category.
2. Using the integers as numerical input may imply an order and meaningful distances between categories. However, these numbers are only labels, so category 3 is not "greater than" category 1.
3. One-hot encoding gives each category its own binary indicator, avoiding an artificial numerical order. Different categories are equally distant under Euclidean distance.

AI used for this answer: Yes

If yes, how: using AI to search for relevant references and correct grammatical errors in my writing.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.LabelBinarizer.html

https://scikit-learn.org/stable/modules/preprocessing.html#encoding-categorical-features

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